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Bounded Relational Dynamics and the Structural Necessity of Born–Infeld Geometry

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Abstract. We investigate how spacetime geometry and non-linear field dynamics can arise as equilibrium descriptions of relational systems subject to a finite maximal relaxation flux. We show that the existence of a universal bound on admissible relational gradients excludes purely quadratic effective actions and uniquely enforces a Born–Infeld-type structure as the minimal local representation compatible with saturation. Starting from a weighted relational Laplacian with irreversible relaxation, we derive an effective continuum description in which the metric tensor emerges from the principal symbol of the operator, while antisymmetric perturbations enter as an effective gauge field strength. In homogeneous regimes, the bounded-relaxation constraint dynamically selects flat spacetime with pseudo-Riemannian signature $(-+++)$ as a stable equilibrium. When homogeneity is broken by a localized stationary obstruction, the same mechanism yields the Schwarzschild geometry as the universal effective exterior solution. Horizon formation corresponds to saturation of admissible relational flux and to a loss of projectability of the continuum description rather than to a physical singularity. These results provide an operator-based unification of Lorentzian geometry, Born–Infeld electrodynamics, and horizon formation as consequences of bounded relational dynamics.

PACS. 04.20.-q Classical general relativity – 11.10.Lm Nonlinear or nonlocal theories and models – 03.50.Kk Field theories in dimensions other than four

1 Introduction

A growing body of work explores the possibility that spacetime geometry is not a fundamental structure but an effective description emerging from more primitive relational or operator-based systems. In several approaches, discrete relational data, weighted graphs, or spectral operators are shown to admit continuum limits in which geometric notions such as distance, dimension, and curvature arise as derived quantities rather than postulated ones. In particular, it has been demonstrated that, under mild regularity and density assumptions, the continuum limit of a relational Laplacian converges to a second-order elliptic operator whose principal symbol defines an effective metric tensor [5].

While such results establish that effective spacetime geometry can emerge from non-geometric relational structures, they leave open a central question: why are only very specific geometries observed? From a purely mathematical perspective, a large class of elliptic operators and corresponding metrics are admissible. However, not all such operators necessarily correspond to physically meaningful or operationally accessible descriptions. Recent analyses have emphasized that effective geometric descriptions may fail when the underlying relational structure admits non-injective or degenerate projections, leading to a loss of observable factorization and to intrinsic limits of spacetime representability [6].

This work addresses a complementary and more restrictive problem. Rather than asking how geometry can emerge, we ask which effective geometries are *dynamically admissible* once minimal physical constraints are imposed on the underlying relational dynamics. Specifically, we consider relational systems whose effective continuum descriptions admit a finite maximal propagation or relaxation speed. In relational frameworks where effective observables arise through generally non-injective projections of an underlying configuration space, the admissible relaxation flux cannot grow without bound. Non-injective projection implies that multiple underlying configurations may correspond to the same effective state, so that relaxation processes must remain compatible with a bounded descriptive flux. Such a bound is required for causal consistency and excludes purely quadratic actions, which allow arbitrarily large gradients and unbounded fluxes.

We show that imposing bounded flux propagation uniquely constrains the form of any local effective functional governing the continuum limit. Under mild assumptions of locality, smoothness, bounded relaxation flux, and absence of additional microscopic scales, the resulting effective description must take a Born–Infeld–type form. This structure is not introduced as a phenomenological modification, but arises as the minimal nonlinear completion compatible with bounded relaxation flux.

Building on this result, we demonstrate that flux saturation does more than regularize the effective theory. It dynamically selects a restricted class of admissible relational Laplacians whose continuum limits define stable and physically meaningful metrics. In homogeneous regimes, this selection leads uniquely to flat spacetime. In the presence of a localized and stationary obstruction, the same mechanism yields the Schwarzschild geometry as the universal effective solution, without postulating any independent metric dynamics or field equations.

Within this framework, horizons are interpreted as loci where flux saturation renders the effective operator degenerate, signaling a loss of projectability rather than a physical singularity. This interpretation aligns naturally with structural analyses of non-injective projections and clarifies the operational meaning of strong-field regimes.

The paper is organized as follows. In Section 2, we establish the necessity of a Born–Infeld–type structure from bounded flux propagation. Section 4 recalls the emergence of effective continuum operators from relational Laplacians. Section 5 discusses metric reconstruction from operator symbols. Sections 6 and 7 derive flat and Schwarzschild geometries as dynamically selected solutions. Section 8 analyzes horizon formation as operator saturation. We conclude with a discussion of scope and limitations.

2 Bounded relaxation and effective action

We consider effective continuum descriptions arising from relational systems whose microscopic dynamics admits a finite maximal transport or relaxation flux. Such a bound is a minimal structural requirement: the underlying relational substrate cannot sustain arbitrarily large effective transport densities without losing projectability.

In a continuum regime, the dynamics of slowly varying configurations can be encoded in a local action functional

$$S = \int d^4x\, \mathcal{L}, \tag{1}$$

where $\mathcal{L}$ depends on the effective fields and their first derivatives. From the perspective of field theory, the central question is therefore the following: which local Lagrangian densities are compatible with a finite maximal transport capacity?

A purely quadratic action, such as the Maxwell form

$$\mathcal{L}_{\text{quad}} \propto F_{\mu\nu}F^{\mu\nu}, \tag{2}$$

allows for arbitrarily large field invariants. In such a linear theory, the constitutive relation $D^{\mu\nu} = \partial\mathcal{L}/\partial F_{\mu\nu}$ remains unbounded. Arbitrarily large field invariants therefore correspond to proportionally large effective flux densities, without intrinsic saturation. This behavior is incompatible with the existence of a finite maximal transport capacity at the microscopic relational level.

Imposing bounded propagation must therefore be understood as imposing bounded constitutive response. The effective Lagrangian density must satisfy three minimal conditions: (i) it reduces to a quadratic form in the weak-field limit, (ii) it enforces a strict upper bound on admissible field invariants through its constitutive relation, and (iii) it introduces no additional microscopic scales beyond the saturation scale itself. Polynomial extensions of quadratic actions fail to meet these requirements without fine tuning or the introduction of auxiliary parameters.

Under these assumptions, the effective Lagrangian density is uniquely constrained to take a Born–Infeld–type form [1]. Anticipating the operator derivation developed in the following sections, the resulting effective action can be written as

$$S_{\text{eff}} = \beta^2 \int d^4x \left(\sqrt{-\det\!\left(g_{\mu\nu} + \frac{1}{\beta}F_{\mu\nu}\right)} - \sqrt{-\det(g_{\mu\nu})} \right), \tag{3}$$

where $g_{\mu\nu}$ is the effective metric and $F_{\mu\nu}$ an antisymmetric field strength. The parameter β sets the maximal admissible field magnitude and fixes the saturation scale of the theory.

The saturation scale β will be referred to as the bounded-flux (Born–Infeld) scale. It characterizes the maximal admissible relational flux within the validity domain of the continuum description and encodes the intrinsic transport capacity of the underlying substrate.

In the weak-field regime $|F_{\mu\nu}| \ll \beta$, the expansion of Eq. (3) yields

$$\mathcal{L}_{\text{eff}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \mathcal{O}\big(\beta^{-2}\big), \tag{4}$$

showing that standard Maxwell electrodynamics is recovered universally as the low-flux approximation of bounded relaxation. Non-linear corrections become relevant only when the microscopic saturation scale is probed.

Crucially, the Born–Infeld structure is not introduced here as a phenomenological modification of electrodynamics. It arises as the minimal local representation compatible with bounded constitutive response. This statement can be formulated as a structural no-go result: any local effective action admitting a finite maximal flux cannot remain purely quadratic in field gradients. Once locality, Lorentz compatibility, analyticity around the weak-field limit, and the absence of additional microscopic scales are imposed, the Born–Infeld class provides the unique analytic completion ensuring saturation of the constitutive relation.

In the present framework, the parameter β is not free. It is determined by the maximal transport capacity and connectivity density of the underlying relational substrate. Consequently, this scale is universal across all effective sectors emerging from the same operator description. Both geometric and gauge degrees of freedom therefore share a common saturation threshold set by the substrate itself.

The remainder of the paper establishes how the effective action (3) emerges dynamically from a bounded relational Laplacian and how this structure selects physically admissible spacetime geometries.

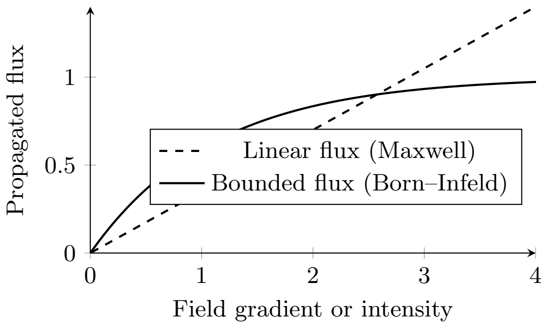

Fig. 1. Comparison between linear and bounded flux propagation. Quadratic actions yield unbounded constitutive response, whereas bounded relaxation enforces intrinsic saturation, naturally selecting a Born–Infeld–type structure.

3 Structural Necessity of Non-Injective Projection

The bounded relaxation principle introduced in Section 2 was formulated as a structural property of the effective description. We now show that this boundedness is not an independent assumption, but a necessary consequence of requiring emergent gauge structure within a relational projection framework.

3.1 Admissible projections and holonomy

Let $\Pi : \Omega \to \mathcal{O}$ denote the projection from the relational configuration space Ω to effective observables $\mathcal{O}$. We assume that admissible projections satisfy locality and compatibility with the relaxation ordering introduced in Section 2.

Proposition 1 *If Π is injective and compatible with relational locality, then the induced connection on the projected fiber bundle has trivial holonomy.*

Proof (Sketch) Injectivity implies that each observable configuration corresponds to a unique relational configuration. Parallel transport of effective states along any closed loop in $\mathcal{O}$ lifts uniquely to a closed loop in Ω. Since no two distinct relational configurations are identified, there exists no non-trivial re-identification symmetry. Therefore the holonomy group reduces to the identity element.

Corollary 1 *An injective admissible projection cannot generate an emergent gauge structure.*

Gauge symmetry requires the existence of non-trivial fiber re-identifications. Trivial holonomy excludes such internal degrees of freedom. Therefore, if electromagnetism is to arise as an effective $U(1)$ fiber symmetry, the projection Π must be non-injective.

Remark on structural necessity. The necessity of non-injective projection invoked here is not a modeling assumption. A general classification of admissible projections shows that any injective projection compatible with relational locality has trivial holonomy and therefore cannot support emergent gauge or curvature structure. Non-injectivity is thus a structural requirement for the emergence of internal fiber symmetries. A complete proof of this classification result is given in [7].

3.2 Finite transport capacity from non-injectivity

Non-injectivity implies that multiple relational configurations are mapped to the same effective observable. Let $\mathcal{E}_y = \Pi^{-1}(y)$ denote the equivalence class associated with an observable $y \in \mathcal{O}$.

Because $\mathcal{E}_y$ is non-trivial, the projection cannot resolve arbitrarily fine variations of relational flux. Transport of relaxation across the projection therefore admits a maximal admissible rate. Beyond this rate, distinct relational configurations collapse into the same effective description, and injectivity fails locally.

Proposition 2 *Non-injective projection implies the existence of a finite maximal effective transport capacity.*

This bound is not imposed externally. It follows from the finite projectability of relational information through Π. The effective field strength cannot grow without limit, since sufficiently large gradients correspond to unresolved relational distinctions.

3.3 Selection of the Born–Infeld completion

Consider a low-gradient regime in which the effective Lagrangian admits a quadratic expansion

$$\mathcal{L}_{\text{eff}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \mathcal{O}(F^4). \tag{5}$$

If transport capacity is finite, the Lagrangian density must remain bounded for large invariants $F_{\mu\nu}F^{\mu\nu}$. Analyticity and Lorentz invariance then restrict the admissible nonlinear completions.

Under these conditions, the minimal deformation that:

1. preserves gauge invariance,
2. reduces to Maxwell theory at low field strength,
3. bounds the field invariants,

is of the Born–Infeld type,

$$\mathcal{L}_{\text{BI}} = \beta^2 \left(1 - \sqrt{1 + \frac{1}{2\beta^2}F_{\mu\nu}F^{\mu\nu} - \frac{1}{16\beta^4}(F_{\mu\nu}\tilde{F}^{\mu\nu})^2}\right), \tag{6}$$

where β sets the saturation scale.

Theorem 1 *If emergent electromagnetism arises from a relational projection that is local, Lorentz-compatible, and non-injective, then the effective nonlinear completion of Maxwell theory must belong to the Born–Infeld class.*

Proof (Sketch) Non-injectivity implies finite transport capacity. Finite capacity requires bounded field invariants. Gauge invariance and analyticity restrict admissible Lorentz scalars to functions of F^2 and $(F\tilde{F})^2$. The Born–Infeld structure is the unique analytic completion that preserves Maxwell behavior at low field strength while ensuring bounded invariants.

3.4 Interpretational consequence

Born–Infeld dynamics is therefore not an arbitrary ultraviolet modification. It is structurally selected once three conditions are imposed:

1. emergent gauge symmetry,
2. relational locality,
3. non-injective projection.

The bounded relaxation principle introduced in Section 2 is thus a necessary consequence of the geometric structure of the projection itself. In the following sections, this structural saturation will be shown to control both spectral stability and horizon formation.

4 Relational graph, Laplacian operator, and relaxation dynamics

We consider an underlying relational system represented by a weighted graph $G = (V, E, w)$, where V denotes a set of nodes, E a set of edges, and $w_{ij} \geq 0$ encodes the strength of the relation between nodes i and j. No embedding in a background spacetime is assumed. The graph is taken as a purely relational object, encoding adjacency and coupling structure between abstract degrees of freedom.

A scalar field ϕ_i is defined on the nodes of the graph. The fundamental operator governing relational variations is the discrete Laplacian, defined by

$$(L\phi)_i = \sum_j w_{ij} \left(\phi_i - \phi_j\right). \tag{7}$$

This operator measures the local mismatch of ϕ with respect to its relational neighborhood and plays the role of a generalized stiffness or connectivity operator. Throughout this work, we assume that the graph is locally finite and that the weights are symmetric, $w_{ij} = w_{ji}$.

The system is assumed to admit an irreversible relaxation dynamics driven by the Laplacian. At the discrete level, this dynamics can be represented schematically as

$$\frac{d\phi_i}{d\tau} = -(L\phi)_i, \tag{8}$$

where τ is a monotonically increasing parameter labeling the progression of the relaxation process. No interpretation of τ as a fundamental physical time is imposed. Instead, it serves as an ordering parameter associated with the irreversible flow toward admissible stationary configurations.

This relaxation dynamics defines a preferred direction of evolution. Configurations evolve toward states that minimize relational gradients, subject to the constraints discussed in Section 2. The existence of such an ordering parameter is sufficient to distinguish one direction of evolution from its reverse, independently of any geometric notion of time. Temporal ordering is thus introduced operationally, as an intrinsic feature of the relaxation process itself.

Stationary configurations satisfy

$$L\phi = 0, \tag{9}$$

and correspond to relational equilibria. More generally, slowly varying configurations describe regimes in which the system admits an effective coarse-grained description. In these regimes, large-scale observables depend primarily on the spectral properties of the Laplacian rather than on the detailed microscopic structure of the graph.

We emphasize that the graph structure introduced here does not imply any fundamental discreteness of physical space or time. It is employed as a relational scaffold allowing the definition of operators and relaxation processes. Different microscopic realizations leading to the same large-scale spectral properties are considered physically equivalent within the scope of the effective description.

In the next section, we recall how, under appropriate density and regularity assumptions, the spectral structure of such relational Laplacians admits a continuum limit. In this limit, the discrete operator converges to a second-order differential operator, providing the bridge toward an effective geometric description.

5 Continuum limit and Born–Infeld action

We briefly recall how an effective continuum description and its associated action emerge from the spectral structure of a bounded relational Laplacian. Detailed proofs of the continuum limit and operator convergence can be found in [5]. Here we summarize only the elements required to establish the effective field-theoretic description in a form sufficient for the subsequent analysis.

We consider a sequence of weighted graphs $G_N = (V_N, E_N, w^{(N)})$ with increasing cardinality $|V_N| \to \infty$. The graphs are assumed to satisfy three structural conditions: (i) uniformly bounded degree growth relative to the intrinsic scaling, (ii) local quasi-isotropy in the sense that weighted second moments of adjacency converge to an isotropic tensor in the low-energy sector, and (iii) sufficient decay of $w_{ij}^{(N)}$ with intrinsic relational distance to ensure locality of the limit operator. No background embedding is assumed. All notions of proximity are defined intrinsically from the graph structure.

Let ϕ_i denote a scalar degree of freedom on the nodes of G_N. The discrete Laplacian acts as

$$(L_N\phi)_i = \sum_j w_{ij}^{(N)} \left(\phi_i - \phi_j\right). \tag{10}$$

Under the above conditions, the family L_N converges in the strong resolvent sense, within a fixed low-energy spectral window, to a second-order differential operator on a smooth manifold $\mathcal{M}$. More precisely, for modes whose eigenvalues remain bounded as $N \to \infty$, the action of L_N on slowly varying configurations converges to

$$L_N \;\longrightarrow\; \mathcal{L} = \nabla_\mu (A^{\mu\nu}(x) \nabla_\nu)\,, \tag{11}$$

where $A^{\mu\nu}(x)$ is a symmetric tensor encoding the local connectivity and stiffness of the underlying relational system.

The convergence holds in the sense that resolvents $(L_N + \lambda I)^{-1}$ converge strongly to $(\mathcal{L} + \lambda I)^{-1}$ for λ in the resolvent set associated with the low-energy sector. This guarantees stability of the principal symbol under bounded perturbations of the microscopic graph structure and ensures that the effective operator is insensitive to microscopic rearrangements that do not alter the spectral window.

The tensor $A^{\mu\nu}(x)$ arises from the second weighted moments of adjacency. Local quasi-isotropy implies that anisotropic corrections remain controlled and contribute only higher-order perturbations to the principal symbol. Consequently, the leading-order propagation properties of continuum modes are robust under bounded microscopic anisotropies.

In admissible continuum regimes, the operator $\mathcal{L}$ is elliptic on spatial slices and admits a well-defined principal symbol,

$$\sigma_2(\mathcal{L})(x, k) = A^{\mu\nu}(x) k_\mu k_\nu, \tag{12}$$

which fully characterizes the leading-order propagation of modes. As shown in [5], this symbol provides a natural, coordinate-independent definition of an effective metric tensor,

$$g^{\mu\nu}(x) \propto A^{\mu\nu}(x), \tag{13}$$

where the proportionality factor reflects a choice of units.

Crucially, the effective metric $g_{\mu\nu}$ is not introduced as an independent dynamical variable. It is a derived object encoding how relational variations propagate in the continuum limit. Distinct microscopic graphs leading to the same principal symbol are therefore indistinguishable at the level of effective geometry. The continuum description is thus defined only up to spectral equivalence within the selected energy window.

When antisymmetric perturbations of the relational connectivity are included, the continuum operator acquires an additional two-form contribution that naturally enters as a field strength $F_{\mu\nu}$. As discussed in Section 2, bounded relaxation constrains the admissible gradients of these perturbations. At the level of the effective continuum description, this constraint is most consistently implemented at the level of the action rather than the equations of motion.

The resulting effective action takes a Born–Infeld form,

$$S_{\text{eff}} = \beta^2 \int d^4x \left(\sqrt{-\det\left(g_{\mu\nu} + \frac{1}{\beta} F_{\mu\nu}\right)} - \sqrt{-\det(g_{\mu\nu})} \right), \tag{14}$$

where β sets the universal maximal admissible relational flux. In the weak-field regime, this action reduces to the Maxwell form, while at large field strengths it enforces saturation of the constitutive response.

The validity of this effective action is restricted to regimes in which the spectrum of L_N admits a stable low-energy sector and the bounded-flux condition remains satisfied. Outside this spectral window, strong-resolvent convergence no longer holds, the principal symbol ceases to provide a faithful description, and geometric or field-theoretic notions lose operational meaning.

In the following section, we show that the bounded-relaxation constraint further restricts the admissible forms of $A^{\mu\nu}(x)$. In homogeneous regimes, these restrictions uniquely select a flat effective geometry with a specific signature.

6 Emergence of Minkowski Spacetime: Spectral Stability and Signature Selection

We now demonstrate that the class of homogeneous and isotropic relaxation regimes uniquely selects a flat pseudo-Riemannian metric. In this framework, homogeneity is defined spectrally: the low-energy sector of the relational Laplacian $\mathcal{L}$ is invariant under the translation group, implying that the effective tensor $A^{\mu\nu}(x)$ reduces to a constant matrix $A_0^{\mu\nu}$.

The selection of the metric signature follows from a structural compatibility requirement between irreversible relaxation, bounded constitutive response, and spectral stability of the effective operator. Let τ denote the monotonic ordering parameter associated with relational relaxation. The effective continuum dynamics must admit a well-posed initial value problem relative to τ in the regime where the projection remains local and injective.

The bounded-flux condition established in Section 2 excludes purely diffusive effective descriptions. A parabolic operator would imply instantaneous spreading of perturbations across the relational network, as is characteristic of elliptic spatial operators coupled to first-order diffusion in τ. Such behavior is incompatible with a finite maximal transport capacity, since it destroys any notion of a finite characteristic cone for effective propagation.

In the homogeneous regime, the principal symbol of the effective operator is determined by $A_0^{\mu\nu}$. Denoting k_0 the frequency conjugate to τ and $\mathbf{k}$ the spatial momenta, spectral stability requires that the dispersion relation define a hyperbolic operator in the sense of Hörmander [2]. Hyperbolicity ensures the existence of a finite characteristic cone and a well-posed Cauchy problem for effective continuum modes.

A purely Riemannian signature $(++++)$ would render the principal symbol positive definite. The corresponding operator is elliptic, not hyperbolic. No finite propagation cone exists in this case, and perturbations spread instantaneously in the effective description, violating the bounded-transport requirement.

Conversely, a signature with more than one time-like direction leads to ultra-hyperbolic equations. Such operators generically fail to admit a globally well-posed initial value problem and are unstable under small perturbations of initial data [3]. They therefore cannot provide a spectrally stable fixed point of the relaxation dynamics.

The only configuration compatible with (i) hyperbolicity of the principal symbol, (ii) existence of a finite characteristic cone, (iii) monotonic ordering along τ, and (iv) bounded constitutive response, is the case in which the ordering direction carries a sign opposite to the spatial directions. After normalization of the maximal relaxation speed to $c = 1$, the effective metric is dynamically fixed to

$$g_{\mu\nu} = \eta_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1). \qquad (15)$$

Lorentz invariance thus emerges as a symmetry of the homogeneous spectral fixed point. The Minkowski metric is not postulated as a background structure. It is the unique effective representation of a maximally symmetric relational equilibrium compatible with bounded relaxation.

The emergence of the $(-+++)$ signature also clarifies the role of Wick rotation. The Euclidean sector corresponds to relational connectivity on a fixed relaxation slice, where the operator is effectively elliptic. The Lorentzian signature encodes the dynamical unfolding of relational structure along the monotonic ordering direction. The null interval $ds^2 = 0$ corresponds to saturation of the effective characteristic cone, where relational transport reaches the maximal rate permitted by the microscopic connectivity of the underlying relational graph. This mechanism parallels the causal saturation encountered in non-linear Born–Infeld-type field theories [4].

In the absence of localized obstructions, all curvature invariants $R_{\mu\nu\rho\sigma}$ vanish identically. The homogeneous relational fixed point therefore corresponds to flat spacetime. In the next section, we show how localized defects break this maximal symmetry and generate the Schwarzschild geometry through the same bounded-saturation mechanism.

7 Localized obstruction and Schwarzschild geometry

We now consider relaxation regimes in which homogeneity is broken by the presence of a localized and stationary obstruction. Operationally, such an obstruction corresponds to a region where the relational connectivity or stiffness of the underlying system is enhanced, thereby constraining the local relaxation dynamics. No assumption is made regarding the microscopic origin of this obstruction. Only its macroscopic effect on the effective operator is considered.

Outside the obstructed region, the system remains stationary and isotropic. The effective continuum operator introduced in Section 5 therefore takes the form

$$\mathcal{L} = \nabla_\mu (A^{\mu\nu}(r)\,\nabla_\nu)\,, \qquad (16)$$

where r denotes the radial distance from the center of the obstruction. Spherical symmetry implies that $A^{\mu\nu}(r)$ depends only on r and decomposes into temporal and spatial components,

$$A^{\mu\nu}(r) = \mathrm{diag}(-A_t(r),\, A_r(r),\, A_\perp(r),\, A_\perp(r))\,. \qquad (17)$$

The sign structure is inherited from the hyperbolic principal symbol selected in Section 6.

In the exterior region, where no additional sources are present, the relaxation dynamics is governed by the homogeneous equation

$$\nabla_\mu (A^{\mu\nu}(r)\,\nabla_\nu \Phi) = 0, \qquad (18)$$

where Φ represents a slowly varying scalar probe of the effective structure. Under stationarity and spherical symmetry, this equation reduces to

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 A_r(r)\frac{d\Phi}{dr}\right) = 0. \qquad (19)$$

This relation expresses conservation of effective flux through spheres of radius r. Its general solution is

$$\Phi(r) = \Phi_0 - \frac{C}{r}, \qquad (20)$$

where C is an integration constant characterizing the strength of the obstruction.

The emergence of the $1/r$ profile is therefore not imposed by symmetry alone. It arises within the class of effective operators already constrained by bounded constitutive response and Lorentz-compatible hyperbolicity. In particular, it holds provided that (i) the continuum projection remains local and injective in the exterior region, (ii) effective isotropy is preserved, and (iii) no additional microscopic scales are introduced beyond the universal saturation scale defined in Section 2. Under these conditions, flux conservation uniquely fixes the radial decay.

Using the identification between the principal symbol of $\mathcal{L}$ and the effective metric established in Section 5, the radial dependence of $A^{\mu\nu}(r)$ translates directly into a position-dependent metric tensor. Up to a choice of coordinates consistent with stationarity and spherical symmetry, the effective line element can be written as

$$ds^2 = -f(r)\,dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \qquad (21)$$

where the lapse function $f(r)$ is determined by the radial profile of the operator coefficients. Matching the weak-obstruction limit with the homogeneous solution of Section 6 fixes

$$f(r) = 1 - \frac{r_s}{r}, \qquad (22)$$

where r_s is a constant proportional to C and measures the integrated flux associated with the obstruction.

The resulting geometry coincides with the Schwarzschild metric. Here, however, it is not obtained from an independent set of metric field equations. It arises as the universal stationary fixed point of a bounded relaxation operator in the presence of a localized and isotropic perturbation. The parameter r_s characterizes the strength of the obstruction in relational units and is identified operationally through its influence on admissible propagation cones and relaxation rates.

This derivation highlights that the Schwarzschild geometry is a universal effective response within a bounded-relaxation framework. Its form is fixed by symmetry, stationarity, hyperbolicity of the principal symbol, and conservation of saturated flux. No independent curvature dynamics or energy–momentum tensor needs to be postulated at the effective level. The geometry encodes how admissible continuum modes propagate when relaxation is constrained by a localized enhancement of relational stiffness.

In the next section, we analyze the behavior of the effective operator near the radius $r = r_s$. We show that this surface corresponds to saturation of the bounded-transport condition, where the principal symbol degenerates and the continuum chart loses projectability, providing an operator-theoretic interpretation of horizons.

8 Horizon as flux saturation and loss of projectability

We now examine the behavior of the effective continuum description in the vicinity of the radius $r = r_s$ identified in Section 7. At this radius, the lapse function $f(r)$ vanishes and the effective geometry develops a horizon in the standard geometric representation. We show that, in the present framework, this phenomenon admits a precise operator-theoretic interpretation in terms of saturation of the principal symbol.

As established in Section 2, the effective action and its associated continuum operator remain valid only as long as the bounded constitutive response is satisfied. In the presence of a localized obstruction, the radial dependence of the operator coefficients $A^{\mu\nu}(r)$ reflects the increasing constraint imposed on relational relaxation. As r decreases, admissible gradients approach the maximal value set by the universal saturation scale β. At $r = r_s$, this bound is reached in the effective description.

At this radius, the principal symbol of the operator

$$\sigma_2(\mathcal{L})(x, k) = A^{\mu\nu}(r)\, k_\mu k_\nu$$

becomes degenerate. Equivalently, the determinant of $A^{\mu\nu}(r)$ vanishes at $r = r_s$. The operator therefore undergoes a local change of type. Hyperbolicity, which guaranteed a well-posed Cauchy problem in the exterior region, is lost at this surface.

More precisely, hyperbolicity of the effective operator is equivalent to the non-degeneracy and Lorentzian signature of its principal symbol. As long as the quadratic form

$$A^{\mu\nu}(r)\, k_\mu k_\nu$$

has full rank and one negative eigenvalue, the operator defines a local characteristic cone and the associated Cauchy problem is well posed. Degeneracy at $r = r_s$ corresponds to a loss of rank of the quadratic form, implying the collapse of the characteristic cone and a local change of operator type.

In particular, the vanishing of $\det A^{\mu\nu}(r)$ does not imply any divergence of curvature invariants constructed from the effective metric. The breakdown occurs at the level of the principal symbol and of the hyperbolic structure, not at the level of curvature singularities. The horizon therefore signals a failure of continuum projectability rather than the emergence of a geometric divergence.

This loss of hyperbolicity implies that the effective continuum description can no longer sustain propagating modes with a finite characteristic cone. The Cauchy problem for effective fields ceases to be well posed at $r = r_s$. Importantly, this degeneracy does not signal a singularity of the underlying relational substrate. It indicates only the breakdown of the continuum chart obtained through projection.

From the perspective of the effective action, the horizon corresponds to the locus where Born–Infeld saturation becomes operative in the constitutive relation. The non-linear structure of the action enforces a maximal admissible flux density, preventing divergences in field invariants and energy densities. As in standard Born–Infeld electrodynamics, saturation regularizes quantities that would otherwise diverge in a purely quadratic theory.

From the viewpoint of the relational dynamics, the horizon separates two distinct regimes of projectability. For $r > r_s$, the relaxation dynamics admits a locally injective projection into a continuum description with well-defined geometric observables. At $r = r_s$, distinct microscopic configurations begin to map to identical effective operator data. The projection from relational states to spacetime observables becomes intrinsically non-injective.

A quantitative measure of this degeneracy can be expressed through a projection entropy defined by the multiplicity of underlying configurations compatible with a given effective observable:

$$S_\Pi = -\sum_{o \in O} \mu(\Pi^{-1}(o)) \log \mu(\Pi^{-1}(o)). \tag{23}$$

Horizon formation corresponds to the regime in which the growth of $\Pi^{-1}(o)$ limits the local resolvability of continuum observables.

This interpretation is consistent with structural analyses of non-injective effective descriptions, in which loss of observable factorization reflects a limitation of the effective representation rather than a breakdown of the microscopic dynamics [6]. In the present framework, the horizon is precisely the surface where bounded transport and hyperbolicity cease to be simultaneously compatible.

No extension of the effective spacetime manifold beyond r_s is required for internal consistency. The continuum description is explicitly restricted to regimes in which the principal symbol remains non-degenerate. The horizon marks the boundary of this domain of applicability.

This operator-based interpretation reframes horizons as kinematic consequences of bounded relaxation rather than as geometric singularities. It explains the universality of horizon formation within the class of bounded relational operators and clarifies why attempts to probe beyond the horizon using effective spacetime observables are intrinsically limited by the saturation of the constitutive response.

9 Discussion and conclusion

In this work, we have investigated how effective spacetime geometries and non-linear field dynamics can emerge from a relational system subject to bounded transport. Starting from a weighted relational graph endowed with an irreversible relaxation process, we have shown that minimal structural constraints severely restrict the class of admissible continuum descriptions.

A first central result is that bounded constitutive response excludes purely quadratic effective actions. Once locality, analyticity around the weak-field regime, Lorentz compatibility of the principal symbol, and the absence of additional microscopic scales are imposed, the Born–Infeld class provides the unique analytic completion compatible with saturation. In this sense, Born–Infeld electrodynamics does not appear as a phenomenological deformation of Maxwell theory, but as the minimal local structure enforcing a finite maximal flux. Standard Maxwell dynamics is recovered universally as the weak-field limit of this saturated regime.

Building on established results concerning the continuum limit of relational Laplacians [5], we have shown that bounded relaxation constrains not only the action but also the admissible class of effective differential operators. In the low-energy spectral window where strong-resolvent convergence holds, the principal symbol of the limiting operator defines an effective metric tensor. In homogeneous and quasi-isotropic relaxation regimes, spectral stability and hyperbolicity uniquely select flat spacetime with pseudo-Riemannian signature $(-+++)$. Minkowski geometry thus arises as a dynamically admissible fixed point of bounded relational relaxation rather than as a fundamental background.

When homogeneity is broken by a localized and stationary obstruction, conservation of saturated flux within the hyperbolic operator class generically produces a $1/r$ relaxation profile. Through the operator–metric correspondence, this profile induces the Schwarzschild geometry as the universal effective description of the exterior region. This result follows from symmetry, stationarity, hyperbolicity of the principal symbol, and bounded transport, without postulating independent metric field equations.

Horizons admit a precise operator-theoretic interpretation. They correspond to surfaces where the principal symbol degenerates and hyperbolicity is locally lost. The associated failure of the Cauchy problem signals a breakdown of the effective continuum chart, not a singularity of the underlying relational substrate. This interpretation is consistent with structural analyses of non-injective effective descriptions [6] and clarifies the operational meaning of strong-field regimes.

Saturation regimes also admit a complementary interpretation in terms of projection-limited descriptions. As the constitutive response approaches its maximal value, distinct microscopic relational configurations become indistinguishable at the level of effective operators. Rather than disappearing, unresolved structure is re-encoded into effective geometric or thermodynamic parameters that compensate for the reduced local resolvability of the continuum representation.

The scope of the present work is deliberately limited. We do not propose a complete microscopic theory of gravitation, nor do we specify the fundamental origin of the relational dynamics. Our results apply only within regimes where a low-energy spectral sector exists, strong-resolvent convergence holds, and bounded relaxation is satisfied. Outside this domain, geometric and field-theoretic notions need not remain valid, and no extension of the effective spacetime manifold is implied.

Within these limits, the framework provides a unified structural explanation for the emergence of flat spacetime, Schwarzschild geometry, and horizon formation as dynamically selected effective structures. It suggests that key features traditionally attributed to gravitational dynamics may reflect universal properties of bounded relational transport rather than independent curvature degrees of freedom. Further work will be required to analyze non-stationary regimes, departures from spherical symmetry, and possible phenomenological consequences of this operator-based perspective.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used generative AI tools as an auxiliary support for language refinement, structural clarification, and consistency checking. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the scientific content, interpretations, and conclusions of the published article.

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